

EdgeMonitor Pilot Incident and Validation Notes

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Project: EdgeMonitor Gateway Pilot

Document type: Incident and validation notes

1. Purpose

This document records early validation findings from the EdgeMonitor pilot.

It should be used as project context when generating regression tests and risk-based test plans.

2. Incident INC-EM-001: Duplicate Telemetry After Cloud Timeout

Severity: High

Status: Fixed

Related requirements:

- REQ-EM-005;
- REQ-EM-006.

Description

During a cloud timeout test, the cloud simulator accepted part of a telemetry batch, but the gateway retried the entire batch after the timeout.

The simulator detected duplicate event_id values for 12 telemetry records.

Root Cause

The forwarder did not reconcile partial success after timeout recovery.

Fix

The forwarder now uses event_id as an idempotency key and reconciles accepted records before retrying.

Required Regression Tests

- replay buffered telemetry after partial timeout;
- verify duplicate records are not accepted;
- verify retry_count is updated only for pending records;
- verify forwarding anomaly appears in logs.

3. Incident INC-EM-002: Diagnostic Export Leaked Token-Like Header

Severity: Critical

Status: Fixed

Related requirements:

- REQ-EM-013;
- REQ-EM-017.

Description

A diagnostic export contained a token-like value in a nested JSON field named `cloud_debug_header`.

Root Cause

The redaction utility scanned known top-level keys but did not recursively inspect nested JSON fields.

Fix

Redaction now recursively scans keys and values using secret pattern detection.

Required Regression Tests

- nested JSON token redaction;
- bearer token redaction;
- private key redaction;
- export audit event;
- exported file checksum recording.

4. Incident INC-EM-003: MQTT Device Health Delay

Severity: Medium

Status: Open

Related requirements:

- REQ-EM-007.

Description

During MQTT broker disconnect testing, device health changed to disconnected after 44 seconds instead of the required 30 seconds.

Suspected Cause

The MQTT adapter waits for two missed heartbeat cycles before notifying the Device Health Service.

Required Tests

- MQTT broker disconnect timing;
- Modbus TCP disconnect timing;
- HTTP sensor timeout timing;
- reconnect transition from disconnected to healthy.

5. Validation Priorities

The test plan should prioritize:

1. authentication and role-based access control;
2. local buffering during cloud outage;
3. duplicate telemetry prevention;
4. diagnostic redaction;
5. device health timing;
6. configuration versioning and rollback;
7. buffer capacity alerting.

6. Useful Demo Questions

The project chatbot should be able to answer:

- Which requirements are related to duplicate telemetry prevention?
- Which incident is still open?
- Which regression tests are required for diagnostic redaction?

- What should happen if TLS validation fails?
- Which role can roll back configuration?
- What are the highest-priority test areas?